

Flu Forecasting Project DRAFT

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Abstract

Each year, influenza epidemics result in significant morbidity, mortality, and cause adverse effects to the economy. Predicting annual influenza outbreaks can prepare jurisdictions to reduce detrimental outcomes through targeted vaccinations, health care preparations, and other programs to reduce exposure. Since probability of exposure among individuals is enhanced through increased human density, we used incidence data from the Southern hemisphere during the winter to predict future influenza rates in the Northern hemisphere. The model predicted a 30% increase in the winter of 2025 in North America which closely tracked actual incidence. We maintain models populated with data from other jurisdictions can be useful to predict relative incidence and be used to more effectively allocate resources from which to protect public health.

1 Introduction

Annual influenza outbreaks cause considerable morbidity, mortality, and economic effects. There is also widespread concern amongst public health professionals regarding the deadly potential of a future influenza pandemic (Doshi 2008). Pathogenic airborne viruses increase transmission proportionally within higher areas of human density. Such increases occur annually during winter in temperate regions when people spend greater time indoors (Simonsen 1999). Tempo-spatial patterns can be useful in predicting influenza outbreaks (Onozuka and Akihito 2008, Shaman and Karspeck 2012) and can be used in modeling approaches to better prepare certain jurisdictions to focus public health resources.

The strength of an influenza outbreak can be dependent on a variety of factors such as the dominant strain(s) of the virus $H(i)N(j)$, efficacy of the vaccine, and proportion of population that received a timely vaccination. Vaccine efficacy in the U.S. relies on the accuracy of the CDC to forecasts the dominant viral strains likely to infect humans come the flu season. Vaccine creation is done by analyzing the current dominant strains effecting humans in countries where and test how the mutate. This does vary year-to-year but has had a relatively small variance in the past 10 years with a 36(9)% estimated reduction in infection. Rather than studying the viral properties which has already shown to be very challenging with vaccine efficacy never exceeding 50%, we propose countries in the southern hemisphere can act as confounding variables where the necessary viral properties and human behavior are hidden in severity rates from different countries.

We first present a subset of countries whose influenza rates may be heavily related to the US, and feed that data into a model designed to predict broad future influenza incidence in the United States with roughly $\frac{1}{2}$ year forecasting power. Long range influenza forecasts are scarce with the current “state-of-the-art” forecasts provided by the CDC (Flu-Site, find better citation and explanation of their work) are done using an ensemble of models that produce forecasts and confidence intervals for 1-4 weeks and generally describe the weekly rates quite well. This work attempts to push the forecasting range, sacrificing some week-to-week accuracy, but capture the spirit of the influenza season.

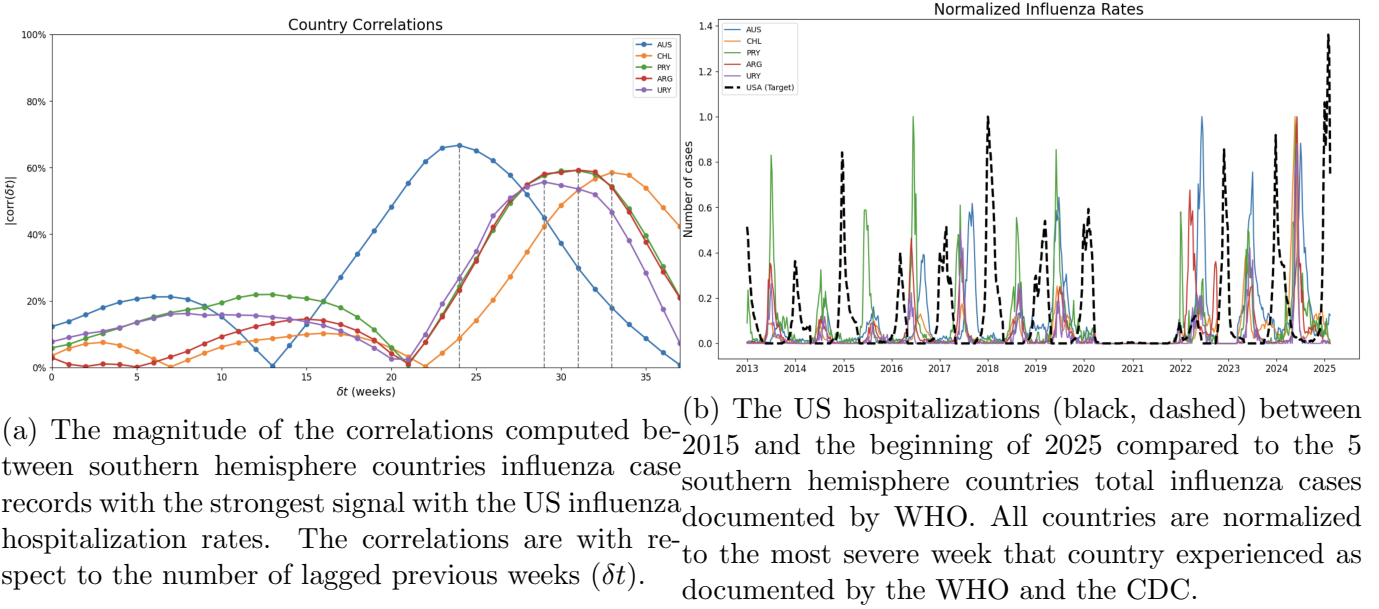


Figure 1: Caption place holder

2 Southern Hemisphere Countries

The nature of the Earth’s tilt, with respect to its orbit around the Sun, provides a phase difference for the seasons of northern and southern hemisphere locations making winters $\sim \frac{1}{2}$ years apart. Given the ease people travel the globe, we assume the viral strains dominating the infections in the southern hemisphere will be related to the strains dominating the northern hemisphere and are not isolated in a way that would cause significant deviations in mutations.

A country’s size and latitude will generally dictate the significance of season change, so not every country in the southern hemisphere will have a winter to observe a flu season. We select a list of relevant “candidates” by borrowing influenza records from the WHO and compute their lagged correlations with the U.S. hospitalizations records given by the CDC (specifically this is the flu-surv dataset which represents $\sim 10\%$ of the population)¹. Specifically, we compute

$$C_{\alpha}(\delta t) = \frac{\sum_t (x_t^{US} - \bar{x}^{US})(x_{t-\delta t}^{\alpha} - \bar{x}^{\alpha})}{\sqrt{\sum_t (x_t^{US} - \bar{x}^{US}) \sum_t (x_{t-\delta t}^{\alpha} - \bar{x}^{\alpha})}} \quad (1)$$

where x_t^{α} represents the number of positive cases reported for country α on week t .

Australia, Chile, Paraguay, Argentina, and Uruguay have the clearest signal and we show this in Figure ?? . These countries have clear peaks and allow us $\sim \frac{1}{2}$ year forecast from a given week.

We can see in Figure ?? that there is no single candidate or obvious linear combination of the group that could represent the target. Each candidate has its own peak height and location. There may be a non-linear relationship that can be captured via tuning a neural network.

3 Models

RNNs (Recurrent Neural Networks) were once the state of the art for sequence learning commonly applied in time-series forecasting and text sequence generation (find some citations). They have lost popularity with the emergence of transformer models (citation), but can still be preferable for small datasets with significant temporal dependencies.

¹We could just as well have used WHO data as the target as well, but CDC hospitalizations were chosen to avoid bias from having the entire project built on a single data source.

We employ RNNs to utilize their feedback connections that enable them to learn sequences of data where ordering is important. These have the particular advantage of not needing a parametric description to capture the seasonality, which can be challenging for weekly inputs where the week start date and year start date do not coincide. The input features the 5 southern countries in batches of 26 week long sequences shifted by 21 weeks making the input tensor of shape $[T_{window}, 26, 5]$ where T_{window} is the total weeks included in the fitting procedure.

The model architecture featured in this calculation can be split into a RNN piece and a sequential piece. The RNN piece may feature a single RNN or two RNNs with a linear layer acting as a feature transformation in between. This general architecture has been used in original text sequence generation techniques where the first RNN acts to capture the semantic meaning of the sentence, the linear layer transforms this semantic representation, and the second RNN is used to predict the next word. We borrow this technique to apply to weekly influenza rates where the first RNN captures the behavior the Southern Hemisphere flu behavior, linear layer transforms into a new representation where the second RNN can then apply to predict the United States flu behavior. The sequential layers are used in dimensionality reduction with the hope they capture the remaining behavior. These layers are illustrated in Figure ??.

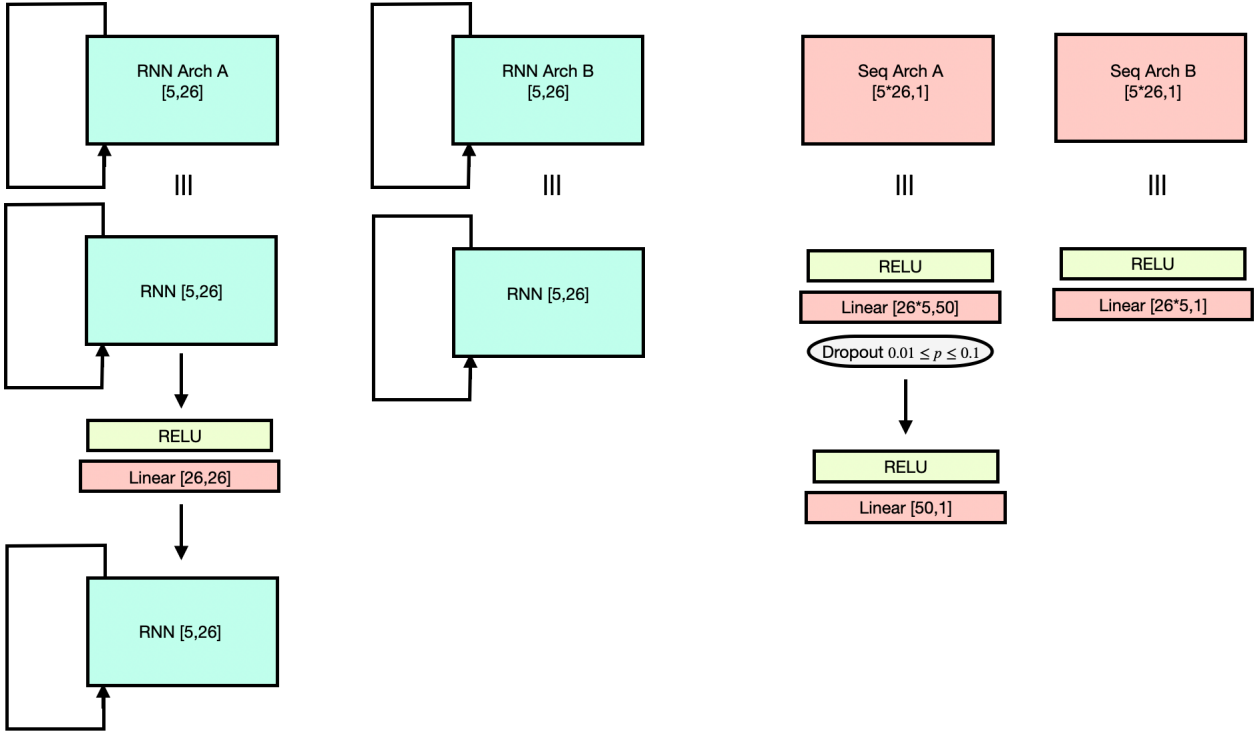
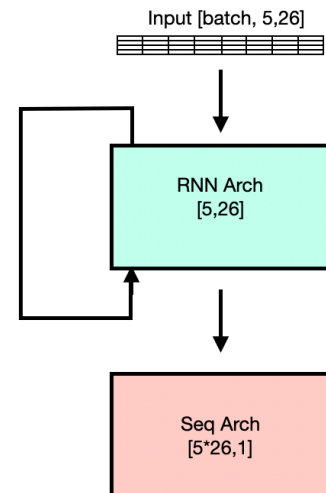


Figure 3: Model archetype composed of two varying combinations of RNNs and sequential layers.

We omit seasons 20-21 and 21-22 due to the damaging effect Covid-19 had on the seasonality. The input during that time period were replaced with zeros to not damage the continuity of the dataset, but we acknowledge we could have just as easily removed that period entirely from training.

We attempt to predict an entire season from previous seasons' data in this so it as natural to use a full season as validation when training the model in an effort to minimize the inevitable overfitting that neural networks are guilty



of. Using weekly data has an advantage over monthly by increasing the sample size, and it is a more natural stratum of influenza records given the rapid change in the season, but does not increase it large enough to avoid overfitting alone. We utilize early stopping and drop out layers in the sequential piece in an effort to combat the overfitting nature of neural networks.

Mean squared error was a loss function for back-propagation, however we invent a slight hybrid to the conventional mean squared error function to evaluate the validation set for early stopping. Our validation metric is the following

$$MSE^{adj} = \sqrt{\left(y_{val}^{max} - y_{val}^{pred,max}\right)^2 + \alpha^2 \left(\left(t_{val}^{max} - t_{val}^{pred,max}\right)^2 + MSE_{train}\right)} \quad (2)$$

where y^{max} is the peak height, t^{max} is the peak location, and the MSE_{train} is the mean squared error of the training set. We invent this metric to ensure the loss function is not penalized for not capturing the exact week-to-week behavior, but rather the “spirit” of the season. The training mean squared error is included to ensure the model has trained sufficiently but is suppressed by a factor of $\alpha = 0.1$ so that the peak height is still the dominant feature when searching for a minima or plateau in the training curve. We suppress the peak location difference as well to keep it on the same relative scale as the peak height.

We found $MSE^{adj} \sim 0.2$ adequately described the validation peaks while simultaneously allowing the model to describe the training set.

The accepted model variations with $\min(MSE^{adj}) \leq 0.2$ are displayed in Figure ?? . These models capture the peak height and even location well but possess spurious movement outside of the flu season and we remove these from the predictions.

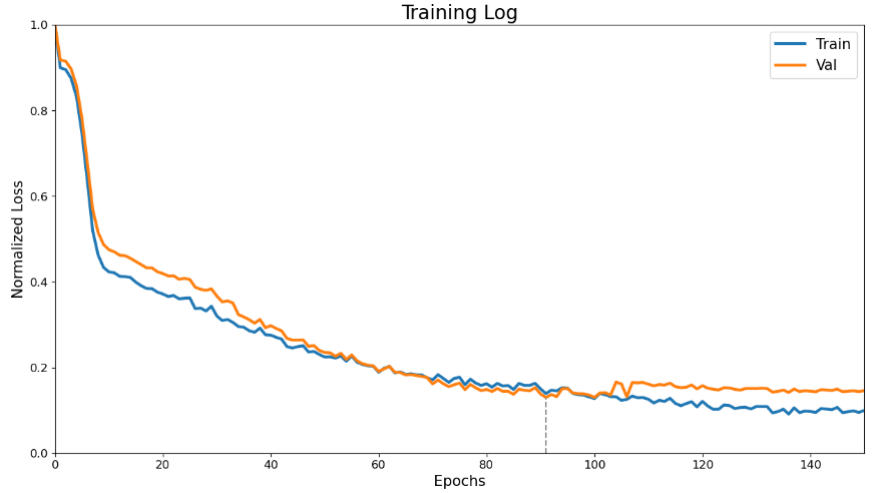


Figure 4: Training curve example for one of the model choices. Training curve is a simple mean-squared-error while validation curve is the hybrid curve explained in this section.

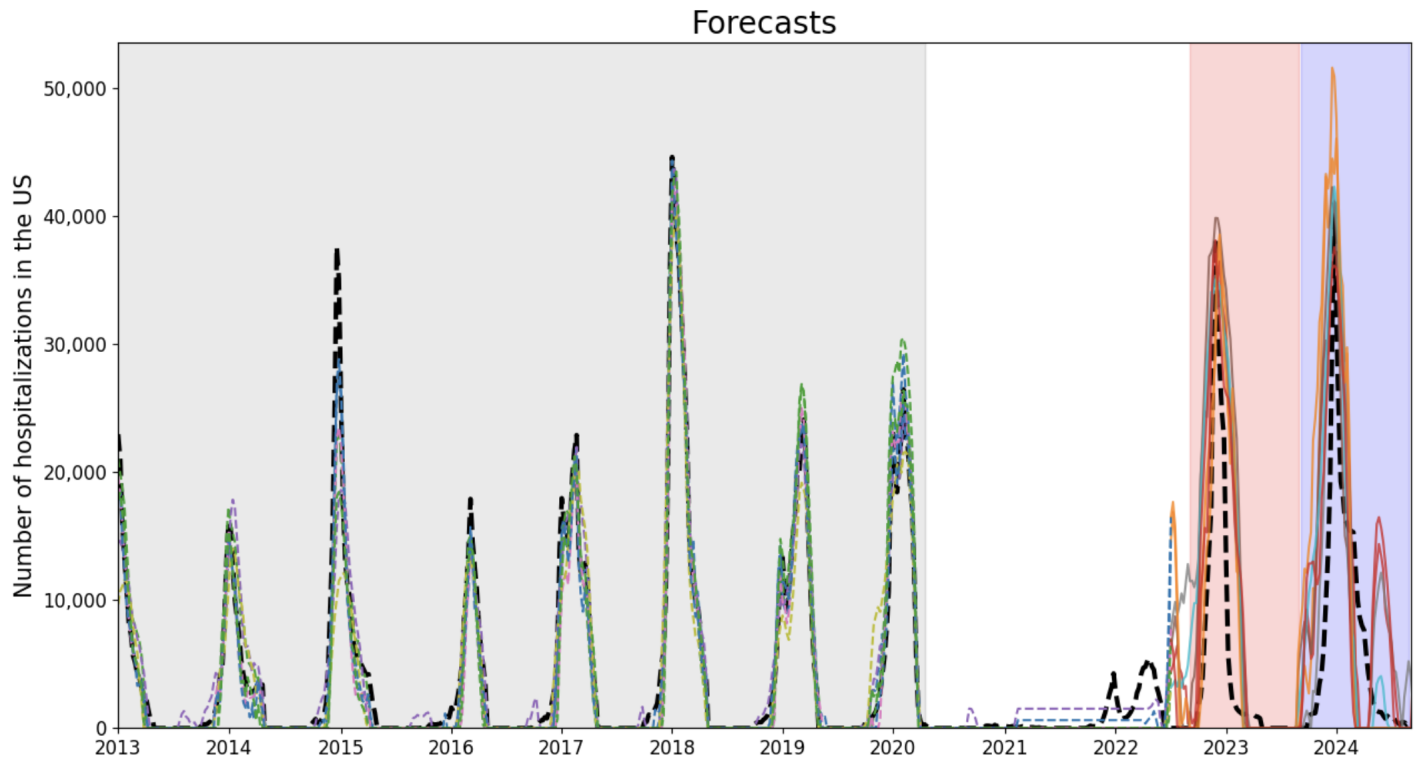


Figure 5: Dashed line represents the target U.S. hospitalization rate. Different colors represent different accepted models with the dashed in the training region and solid lines are outside. Red shaded region includes the validation region, and blue shaded represents the test region.

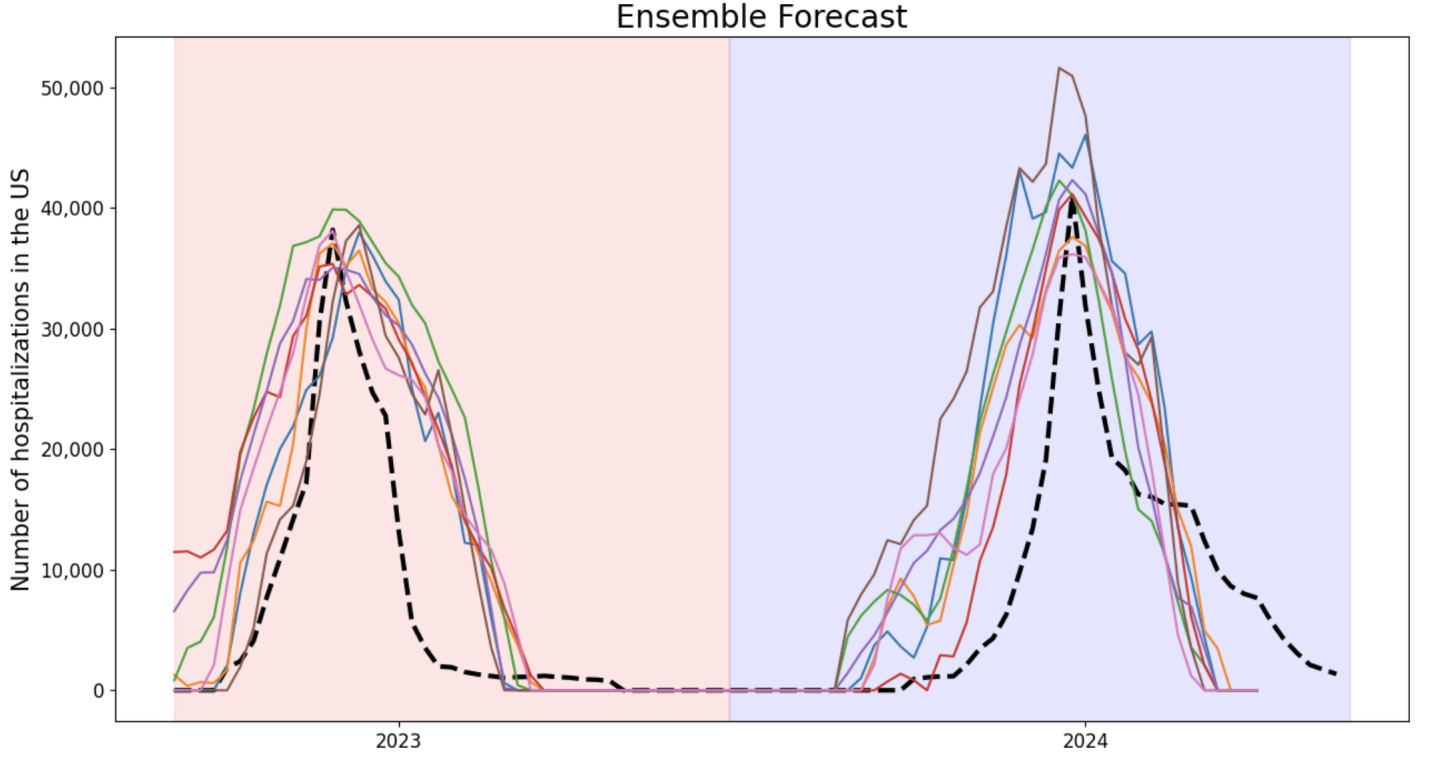


Figure 6

A naive weekly average of the model variation will produce a poor description of the data due to the seasonality where a any small phase difference of two otherwise completely equivalent descriptions would wash away the peak behavior. One must first capture the phase differences and shift accordingly. A parametric solution to combining is the following sum of renormalized hybrid Lorentzian distribution

$$L(t) = \sum_i A_i \frac{\left(\frac{\Gamma_i}{2}\right)^2}{(t - t_i)^2 + \left(\frac{\Gamma_i}{2}\right)^2} \quad (3)$$

where A_i is the peak height, t_i peak location, and Γ_i is the half width. A typical Lorentzian has a defined normalization that we have removed to control the height. All three of the parameters can be computed for each model and their averages and deviations can be used to estimate the model variance. Use a sum due to the double peak nature of some seasons.

Figure ?? demonstrates these parameterizations work remarkably well and each flu season can be reduced to 6 parameters..

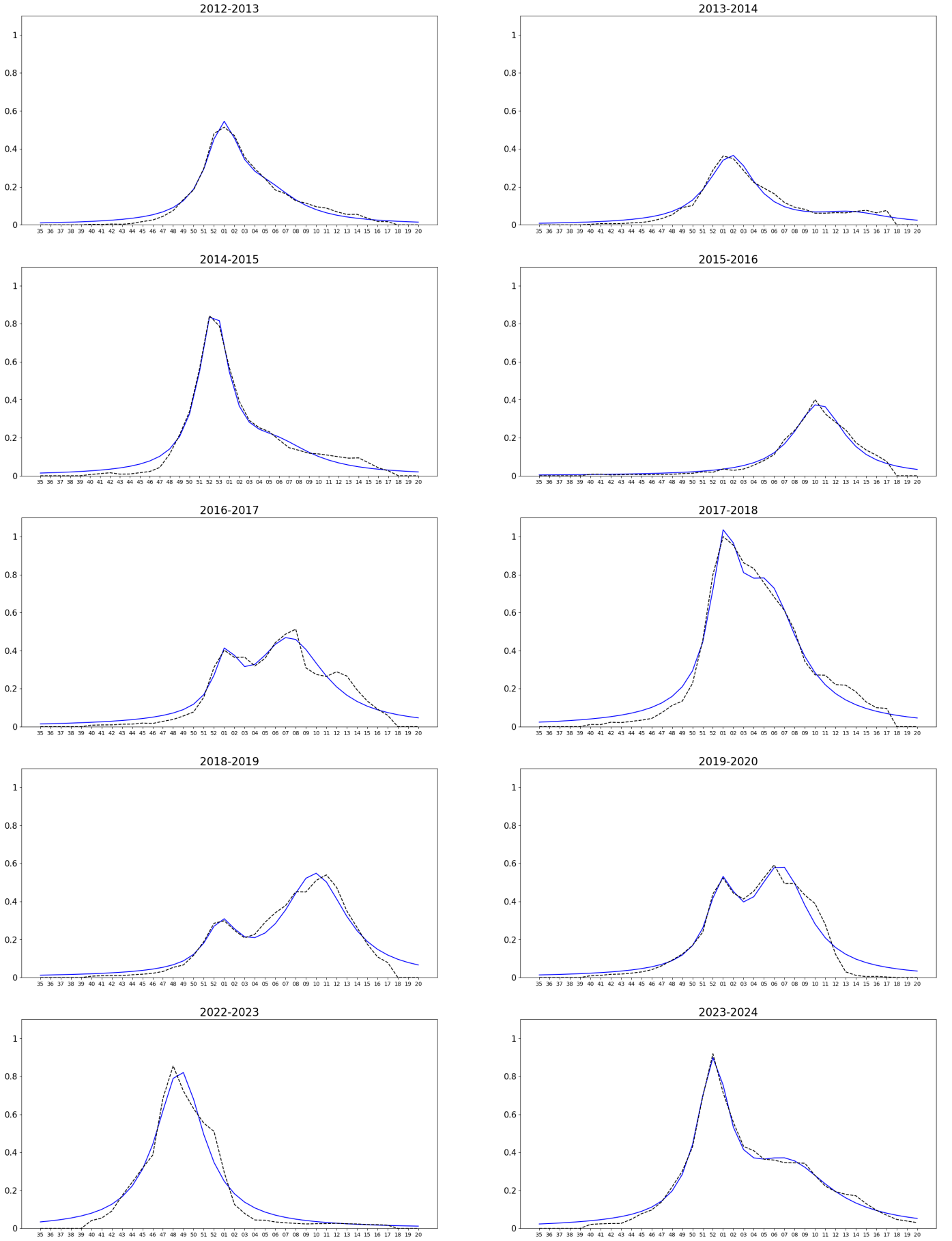


Figure 7: Double Peak fits.

We acknowledge that our data may capture the relative strength of the flu season, but cannot distinguish timing or whether the season is single or double-peaked. We carry forward with a single peak approximation due to the sensitivity of the data.

We note the validation model has over-predicted the width by roughly a factor of 2 which we can account for in the test piece.

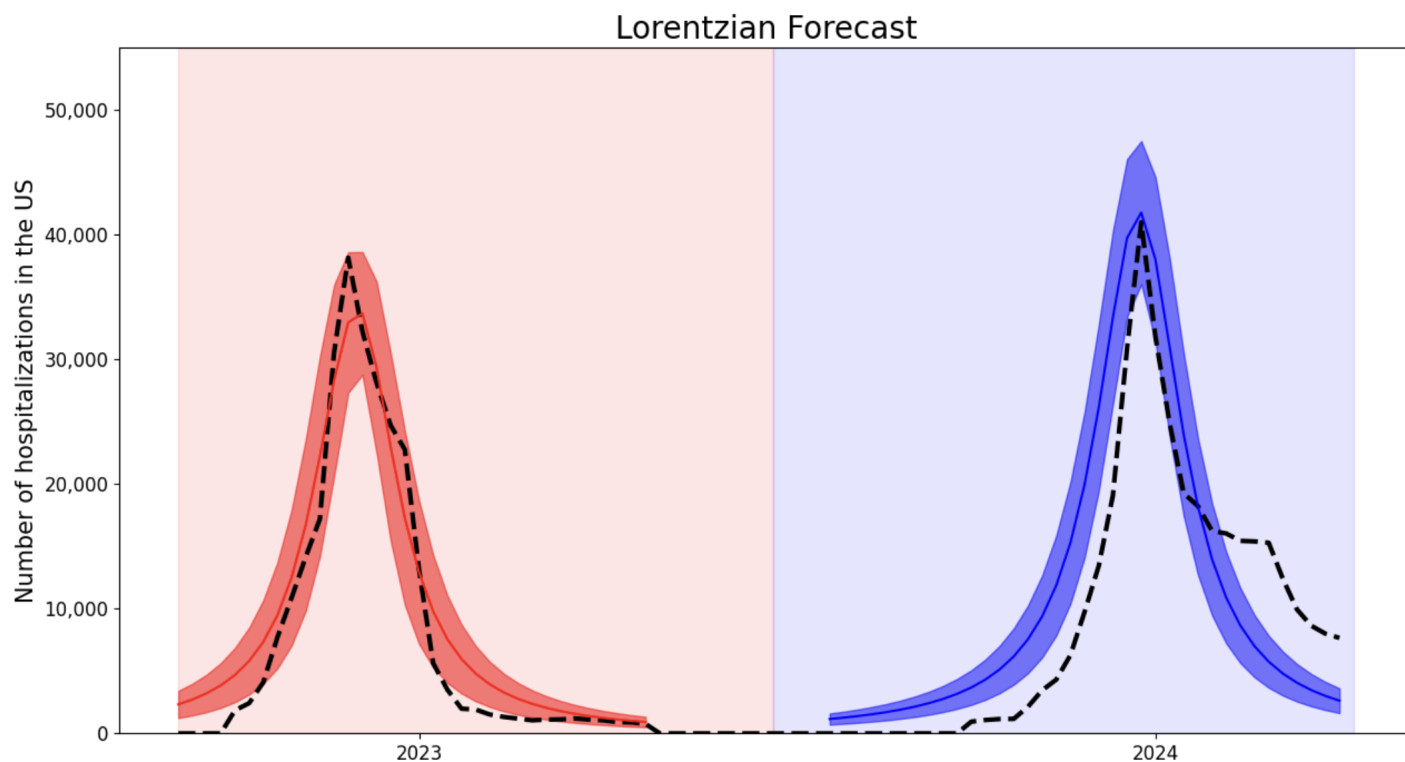


Figure 8

Repeat the same procedures for next season.

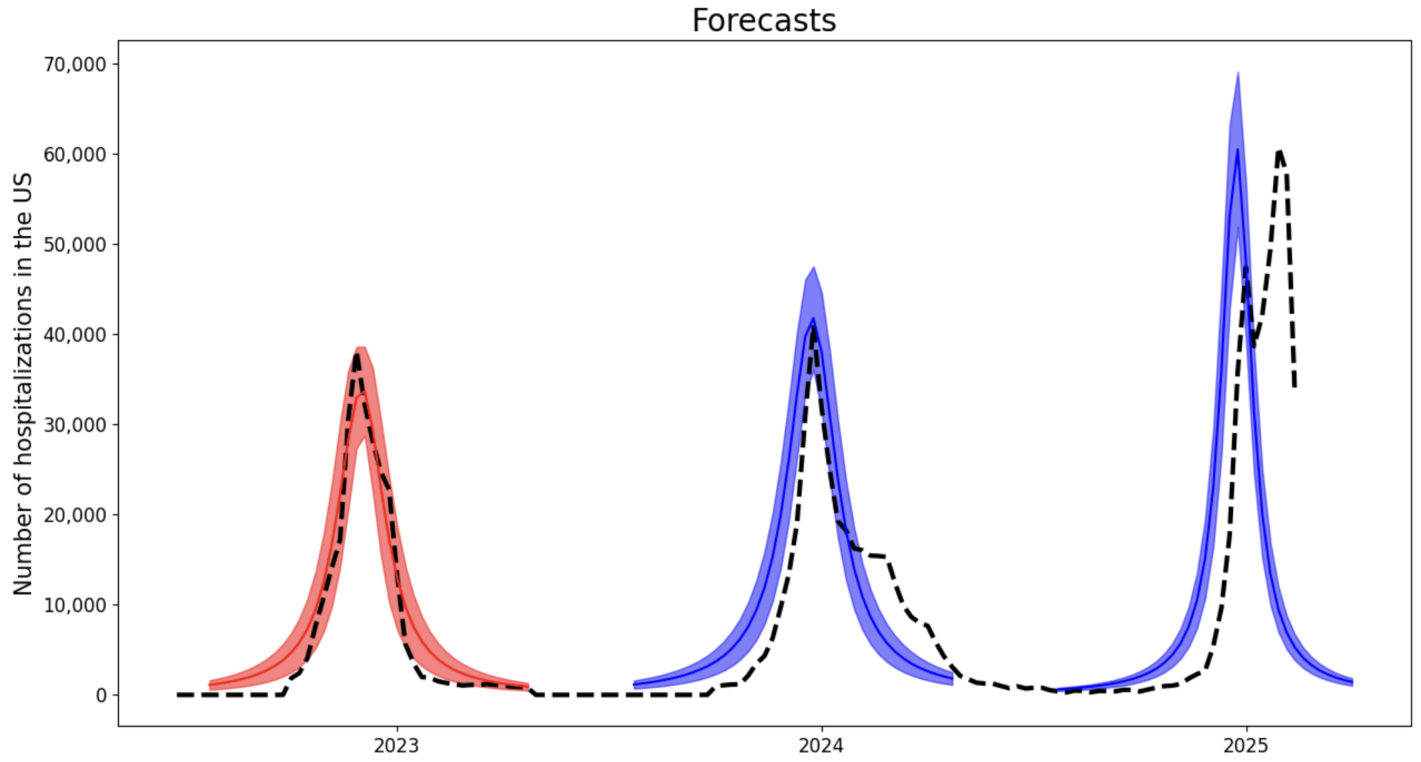


Figure 9

Remark one could in principle use fit parameters as targets and reduce the dimensionality of the problem, however training becomes challenging because each year is reduced to a single data point. Note factors that influence multi peak behavior.

Captures dominant peak behavior but it appears to produce same peak location as previous season. Description also lacks the ability to capture the double peak behavior the flu season experiences.

4 Conclusion